Multi-Target Trajectory Prediction for Drones with Non-Overlapping FoVs

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Abstract

Aerial monitoring and multi-target trajectory prediction(MOT) with fleets of low-cost drones faces two intertwined challenges: each camera captures only a shifting, partial slice of agent motion, and yet predictions must remain globally consistent across areas no drone can currently see. We introduce a centralized multi-agent trajectory-forecasting framework that fuses the masked tracklets each drone transmits with a georegistered semantic map of the entire scene. During training, trajectories falling outside every simulated camera's field of view are hidden from the network's inputs but still supplied as future supervision, compelling the model to exploit static-scene cues—road topology, walkways, obstacles—and internal motion priors whenever dynamic observations disappear. A shared encoder embeds each drone's partial observations alongside its local map patch, a cross-attention fusion module aligns these features against the global map to resolve long-range dependencies, and a probabilistic decoder produces full-horizon position distributions whose aleatoric uncertainties naturally expand once an agent leaves every field of view. Since this is a completely new concept, we've come up with our own benchmarks to verify the efficacy of the model.

1 Introduction

Trajectory prediction. Forecasting how dynamic agents will move through space underpins robust planning for autonomous vehicles, social robots and video-analytics systems. Early deep-learning work such as (Alahi et al., 2016) introduced recurrent models that explicitly pool neighbouring agents' hidden states, allowing predictions that respect collision-avoidance and other social conventions

. Subsequent generative approaches, notably SocialGAN, highlighted the multimodal nature of human motion by producing diverse yet socially acceptable futures through an adversarial framework (Gupta et al., 2018). More recently, modular graph-based encoders such as Trajectron++ have demonstrated how heterogeneous map data and agent-specific dynamics can be fused to output full trajectory distributions suitable for downstream planning(Salzmann et al., 2020)

Multi-target trajectory prediction. When multiple agents interact, joint forecasting must model both individual kinematics and inter-agent dependencies. Probabilistic graph-structured networks like the original Trajectron (Ivanovic and Pavone, 2019) and its successors learn interaction graphs whose edges adapt over time.

These advances have pushed accuracy and uncertainty calibration forward, yet they still assume a single, fixed observer with a complete view of the scene.

To extend coverage beyond a single field of view (FoV), researchers have begun forecasting across camera networks. Multi-Camera Trajectory Forecasting with Trajectory Tensors formalised the task of predicting where, when and in which camera an object would next appear, introducing tensor representations that link coordinate frames across views (Styles et al., 2021).

Follow-up work leveraged predicted futures to improve cross-camera re-identification and multi-camera people tracking, demonstrating that anticipating motion can resolve ID switches at hand-offs between disjoint views(Jeon et al., 2023)

Multi-drone, non-overlapping-FoV, multi-target trajectory prediction. Despite this progress, existing methods assume a very convenient situation of overlapping field of views(FoV), implying a redundancy in drones/cameras. But it is highly likely that situations arise where there are only a small number of drones which are not able to cover the entire scene and have non-overlapping field of views (FoV). Predicting jointly consistent trajectories under these conditions introduces new challenges:

(i) sparse, view-dependent observations with large blind regions; and

(ii) the need to fuse heterogeneous map priors with partial tracklets streamed asynchronously from each drone.

In this paper we extend trajectory prediction to non-overlapping aerial views. We position our work at the intersection of multi-camera forecasting and aerial-robot perception, highlighting the novel combination of

(a) non-overlapping FoVs,

(b) semantic map fusion and

(c) uncertainty-aware forecasting. We propose a framework for multi-target trajectory prediction in a network of non-overlapping, drone cameras. By learning to align each drone's locally observed tracklets with a geo-registered global map and training with hidden-future supervision, our model produces probabilistic, forecasts that remain coherent even when every agent leaves the union of current FoVs. We show that this approach bridges the gap between ground-based multi-camera forecasting and aerial-robot perception.

2 Dataset and methodology

For the problem we used the Stanford Drone Dataset(Arya Shah). In this we simulated multiple drones in the different maps. Different maps having different number of drones. This was done for 2 main reasons:
1. The model should increase the variance (uncertainty in prediction) if the object leaves the FoV.
2. The inference situation should match the training situation. During inference the agents will only be first visible in the drone FoV.

While training, the frames were samples at a subsampling rate of 12. which means we chose every 12th frame, since the fps for the videos are 30hz and sampling at 6 - 12 provides enough time for agents to actually move.
We track a max_agents number of agents for every time frame. For agents that are entirely in the FoV for the entire duration of the input sequence (T_past) we pass the raw trajectories, for targets that leave the FoV, we replace the raw targets with an $unobserved\_mask$, this teaches to model to increase the variance as the prediction gets farther away from the FoV with time. Finally, for agents that cease to exist during the duration of the T_past, we pass in a padding value. We create masks for both, unobserved and not existing agents so that the loss function doesn't penalize them while training.

3 Models

Since there aren't any established benchmarks for this scenario, we have created our own benchmarks. The bench mark consists of a decoder only model with transformers. We're predicting the mean(μ) and log-variance($log\sigma^2$) of the change in x and change in y for the predicted trajectory since the real agents can take any trajectories and this directly models the inherent uncertainty in human and environmental factors. To accurately predict the future positions we have come up with a model that can consider inter-agent interaction. For this, the model has 3 parts:

1. The encoder that's assumed to be in the drone
2. The Graph Attention Network that's assumed to be on the centra serve
3. The decoder, also in the central server.

the inputs to the model while training are of the form.

$$(Batch\_size, num\_agents, seq\_len, 2)$$

2 is for the x, y coordinates of the centre of the bounding boxes of the agents. We're making an important assumption that the tracking is already done.
The encoder only runs attention on the seq_len dimension by reshaping the input tensor to

$$(Batch\_size * num\_agents, seq\_len, 2)$$

while the decoder alternates between

$$(Batch\_size, num\_agents, seq\_len, 2)$$

, and

$$(Batch\_size * seq\_len, num\_agents, 2)$$

This is done because there can be interactions across drones that should be considered and will be in the decoder, and that it's the job of the GNN to decide which agents affect who.
Before feeding in the input, we add the following embeddings:

1. Temporal encoding: this is for the transformer.

2. Spatial encoding: since we're working with delta x and delta y instead of the raw x,y coordinates, we add spatial encoding to the input.

3. label encoding: since there are different types of agents, like bicycles, cars, pedestrians we add a separate embedding for each of these classes to help the model distinguish between them.

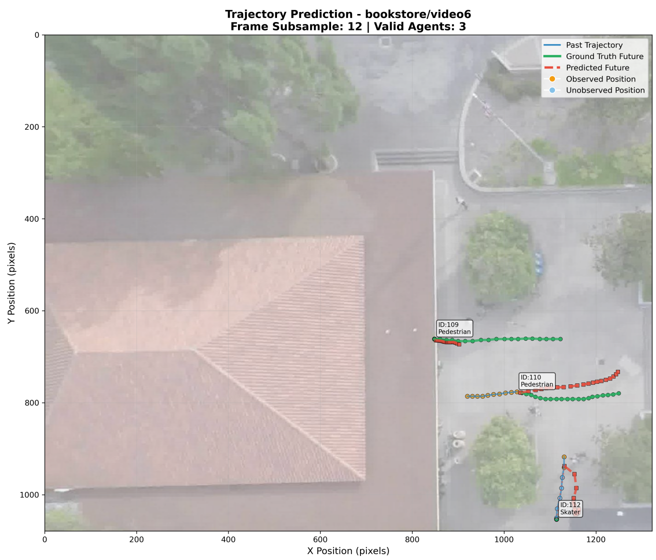
4. gap encoding: Since there are situations when an object leaves the FoV and we add learned unobserved tokens for its input trajectory, we add a gap encoding which counts the number of time steps for which the object is outside the FoV.
For the loss function we're using the Gaussian negative log likelihood. Since it allows us to capture the heteroscedasticity(changing variance basically) and the the continuous nature(as opposed to discrete) of the output.

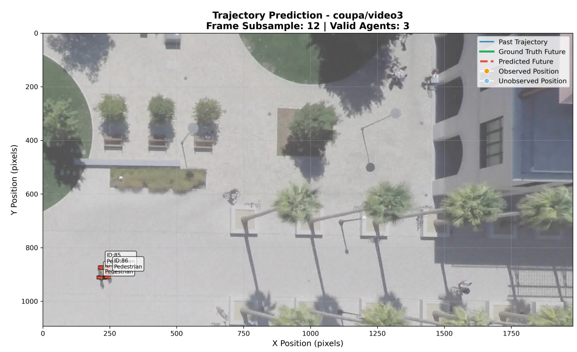
4 Metrics

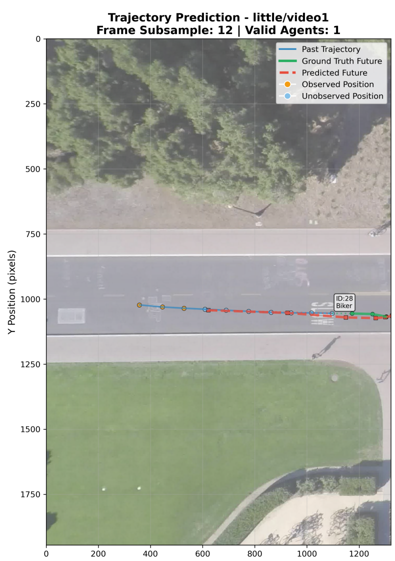
For metrics to evaluate the performance we're using 2 metrics that are usually used together:
1. ADE (Average Displacement Error): Average $L2$ distance between the predicted and observed locations for the intermediate time steps.

2. FDE (Final Displacement Error): The $L2$ distance between the predicted and observed locations for the intermediate time steps.
Usually, to capture the multi-modality (different futures) of trajectory prediction, these metrics are calculated using the k best predictions of the trajectories. Currently, we're limiting it to just singe prediction done using the mean (μ). But this will be expanded in the future.

5 Results







The project is ongoing and there's a lot more work that'll be done in the next couple of weeks. For the baseline, the metrics are:
Val ADE: 107.225
Val FDE: 146.320
for the model with the alternating time and agent dimensions is:
Val ADE: 8.5290
Val FDE: 15.9796
Both of these were trained for 5 epochs.
The model currently has many areas of improvement which involves predicting multiple time steps into the future for the first autoregressive prediction. currently, the model starts predicting right after the targets leave their FoVs, even though that is a part of the input sequence. Additionally the input to the model is highly padded (90% - 95%)since at a given time, very few agents are being tracked and this means that most of the entries in the agent dimension(N, T, 2 (N = agents)) are padded.

6 Future Steps

There are 2 future steps from here:
1. We need to implement the GNN. This will mitigate the massive amount of padding in the model since we'll be able to chose the k most important neighbors to a target.
2. We should feed in the scene information. Possible using a cross-attention model after the GNN layer.

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